

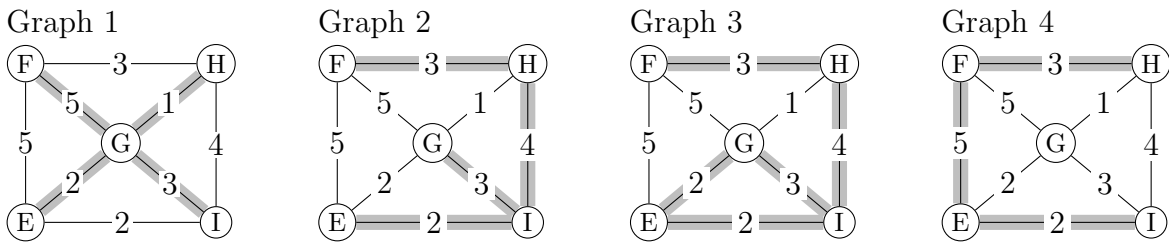
Worksheet: Minimum Cost Spanning Tree

Group Names: _____

1. Find two different spanning trees in the graph below. Draw one on each copy of the graph.



2. Here are four different subgraphs in a weighted graph. Graphs 1 and 2 are spanning trees.



- (a) What is the total cost of the spanning tree in Graph 1? _____
- (b) What is the total cost of the spanning tree in Graph 2? _____
- (c) Which spanning tree has smaller total cost? _____
- (d) Why is the subgraph in Graph 3 not a spanning tree? _____
- (e) Why is the subgraph in Graph 4 not a spanning tree? _____

3. Use Kruskal’s Algorithm to find a minimum cost spanning tree in the following graph. Construct the tree in the second graph. Keep track of the steps of the algorithm in the table below.



Used?	edges	weights

What is the total cost of the spanning tree you found? _____

4. Use Kruskal’s Algorithm to determine a minimum cost spanning tree in the following graph.
- Break ties by choosing the edge that comes earlier in the alphabet.** For convenience, the edges and weights are provided in tabular form, too.



Sorted edges	weight	used?
<i>AB</i>	1	
<i>BF</i>	1	
<i>DE</i>	1	
<i>DG</i>	1	
<i>EG</i>	1	
<i>HI</i>	1	
<i>AD</i>	2	
<i>BE</i>	2	
<i>EH</i>	2	
<i>FI</i>	2	
<i>GH</i>	2	
<i>EF</i>	3	
<i>BC</i>	4	
<i>BD</i>	4	
<i>CF</i>	4	
<i>EI</i>	4	

What is the total cost of the spanning tree you found? _____

5. Why are there instructions about how to break ties? Is this important? What are the consequences?
6. (Bonus) How many edges could a graph on 9 vertices have?